Intelligent Systems Division

NASA Software Engineering Procedural Requirements, Standards, and Related Resources

NASA Procedural Requirements (NPR) Software Engineering Requirements 7150.2D

Engineering requirements for software acquisition, development, maintenance, retirement, operations, and management consistent with the governance model contained in NASA Policy Directive (NPD) 1000.0, NASA Governance and Strategic Management Handbook.

NASA Software Assurance & Software Safety Standard (NASA-STD-8739.8)

The purpose of the Software Assurance and Software Safety Standard is to define the requirements to implement a systematic approach to Software Assurance (SA), software safety, and Independent Verification and Validation (IV&V) for software created, acquired, provided, or maintained by or for NASA. Various personnel in the program, project, or facility, and Safety and Mission Assurance (SMA) organizations can perform the activities required to satisfy these requirements. The Software Assurance and Software Safety Standard provides a basis for personnel to perform software assurance, software safety, and IV&V activities consistently throughout the life of the software, that is, from its conception, through creation to operations and maintenance, and until the software is retired.

Software Formal Inspections Standard (NASA-STD-8739.9)

The Software Formal Inspections Standard is designed to support the inspection process of software developed for NASA. Its goal is to provide a framework and model for an inspection process that will detect and eliminate defects as early as possible in the software lifecycle.

NASA Software Engineering Handbook

This handbook provides users and practitioners with guidance material for implementing the requirements of NPR 7150.2, NASA Software Engineering Requirements. Use of this handbook is intended to provide “best-in-class” guidance for the implementation of safe and reliable software in support of NASA projects. This handbook is a key component of the NASA Software Working

Group's (SWG) implementation of an Agency-wide plan to work toward a continuous and sustained software engineering process and product improvement.

The SWG designed this Handbook for the community that is involved in the acquisition, management, development, assurance, maintenance, and operations of NASA software. Readers can use it to sharpen their skills in specific areas or suggest valuable guidance for others in the NASA software community. Novice and experienced software team members can use the Handbook as an easily accessible reference or manual that captures the broad knowledge base of numerous experts who have extensive experience in all aspects of NASA's software systems.

Secure Coding Portal

This portal provides information to software developers on how to develop code in a secure fashion. It is a single touchpoint for learning about the rules, guidelines, tools, resources, and requirements for coding securely. In addition to the portal, the Secure Coding Newsletter is distributed here.

Ames Software Engineering Procedural Requirements

Ames Procedural Requirements (APR) Software Engineering Requirements 7150.2

This Ames Policy Directive (APD) provides direction for compliance to the NASA Procedural Requirements (NPR) Software Engineering Requirements 7150.2C, defining Ames-specific responsibilities.

NASA Policy Requirements, Directives, and Technical Standards

NASA Online Directives Information System (NODIS)

NASA Policy Requirements (NPRs) and NASA Policy Directives (NPDs) are available here.

NASA Technical Standards

NASA Technical Standards (STDs) are available here.

Related Sites

NASA Technology Transfer Program: NASA Software Catalog

Intelligent Systems Division 2019 CMMI SCAMPI A Appraisal Results (search "National Aeronautics and Space Administration")



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